

Thm: if L is linear and we know L on $\{x_1, \dots, x_n\}$

$\hookrightarrow \text{span}\{x_1, \dots, x_n\} = A \iff L$ is known everywhere in
its domain

• also linearly independent

note: $\{x_1, \dots, x_n\}$ is finite (n elements)

$\text{span}\{x_1, \dots, x_n\}$ is infinite iff the field of scalars is infinite.

Def: A set of vectors $\{x_1, \dots, x_n\} \subset V$ that

- is linearly independent
- spans V

is called a basis for V

Ex: $\{(1,1), (1,-1)\} \subset \mathbb{R}^2$ is a basis

$\{(1,0), (0,1)\} \subset \mathbb{R}^2$

Fact: bases are not unique, infinitely many bases

Thm: if V has a finite basis then

1) all bases of V are finite

2) all bases of V have the same # of elements

Def: if V has a finite basis, we call the size of that set the dimension of V

Ex: $\{(1,1), (1,-1)\}$ is a basis for $\mathbb{R}^2$ we say
 $\dim \mathbb{R}^2 = 2$

Ex: What are some bases for $\mathbb{C}$ (as a vector space over $\mathbb{C}$)? $\{1\}$ or $\{i\}$ cuz of rotation in $\mathbb{C}$